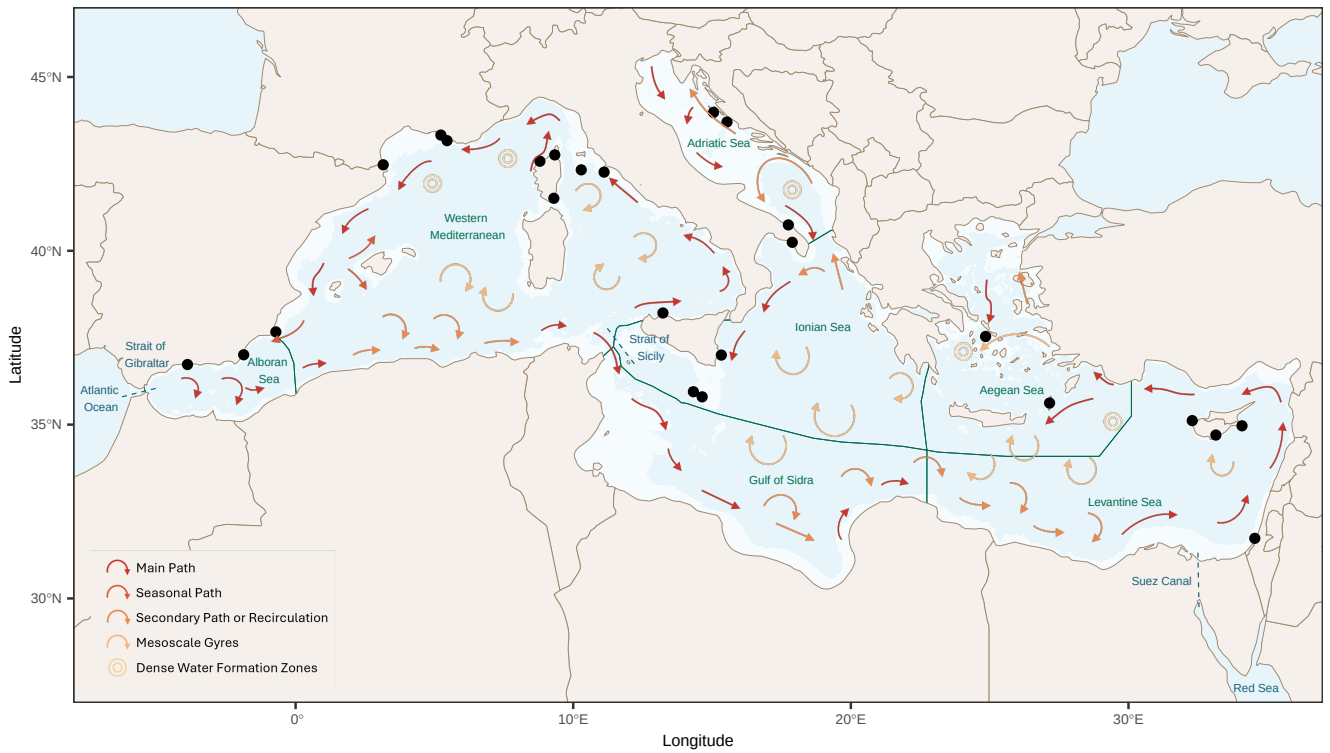
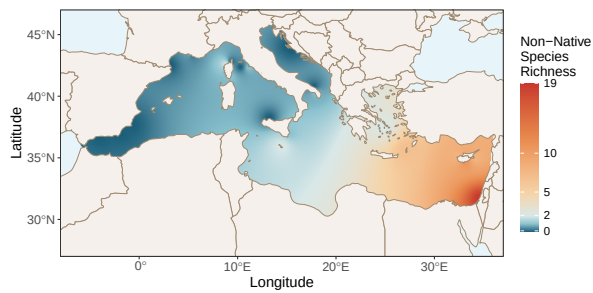


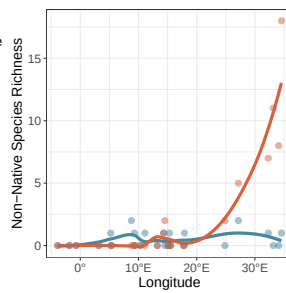
(a)



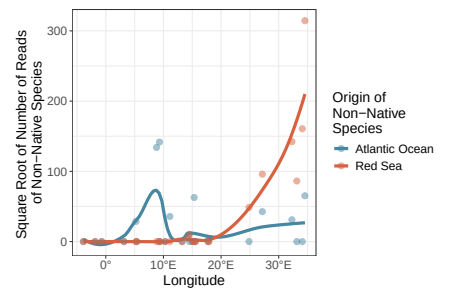
(b)



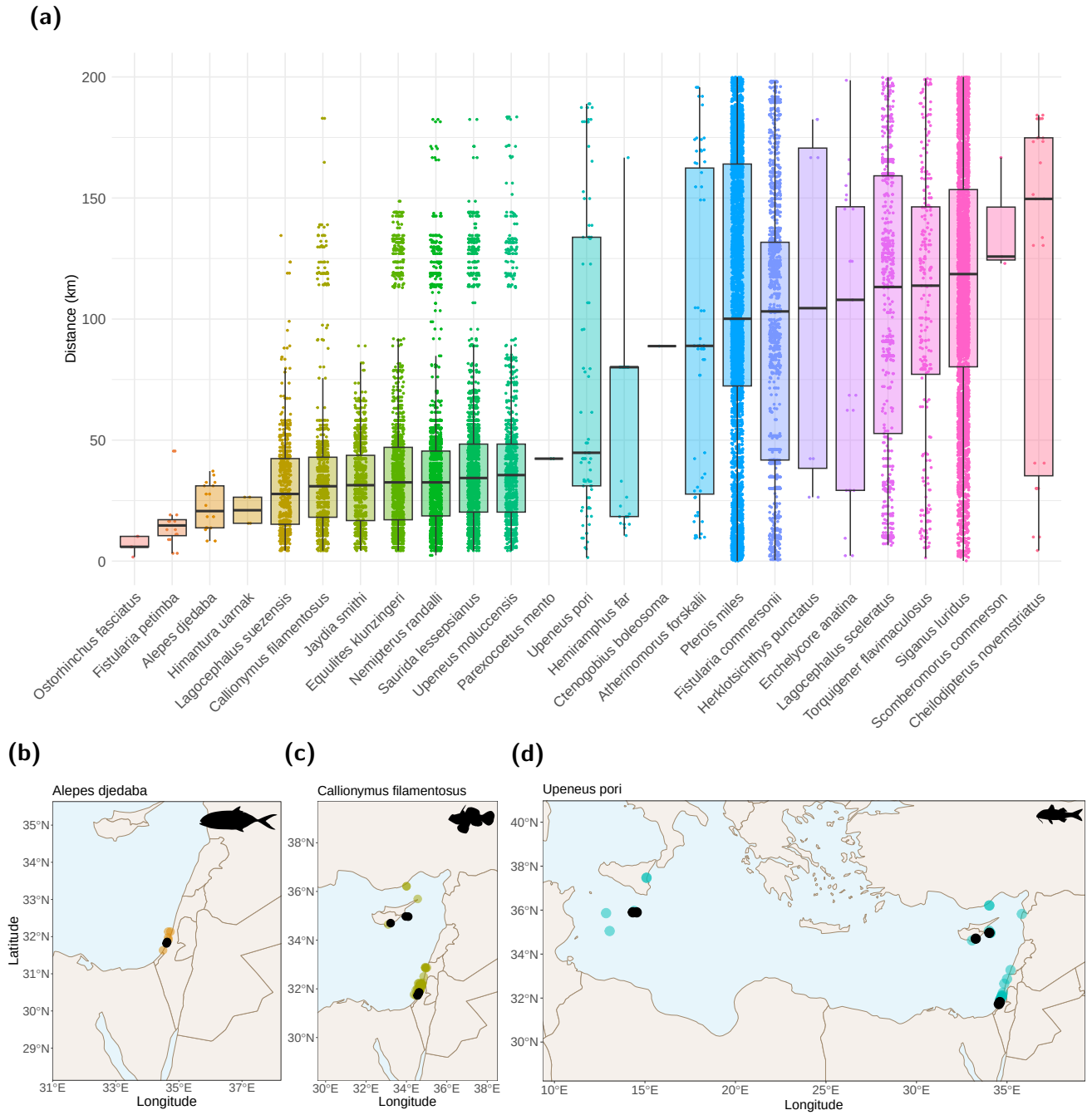
(c)



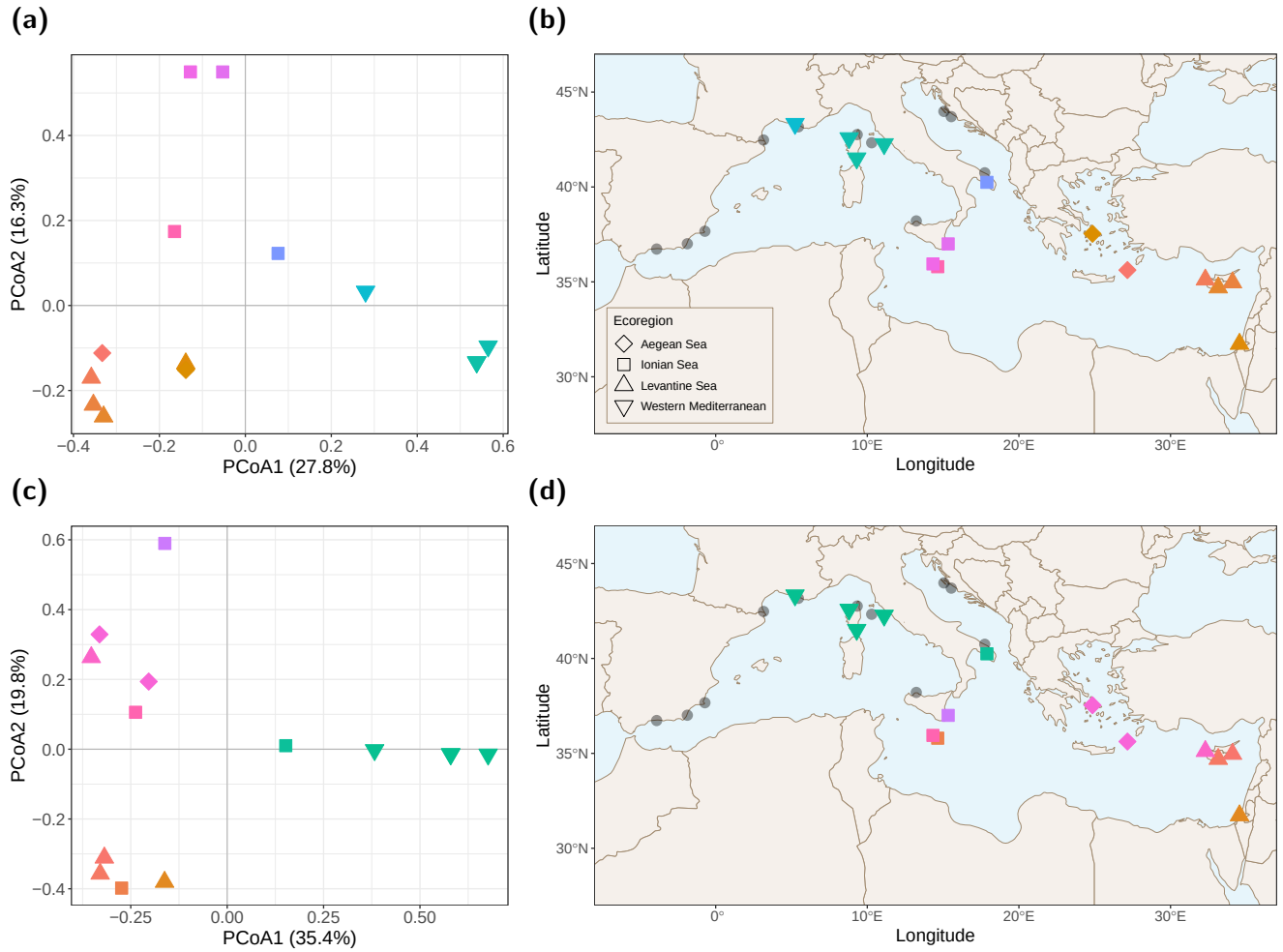
(d)



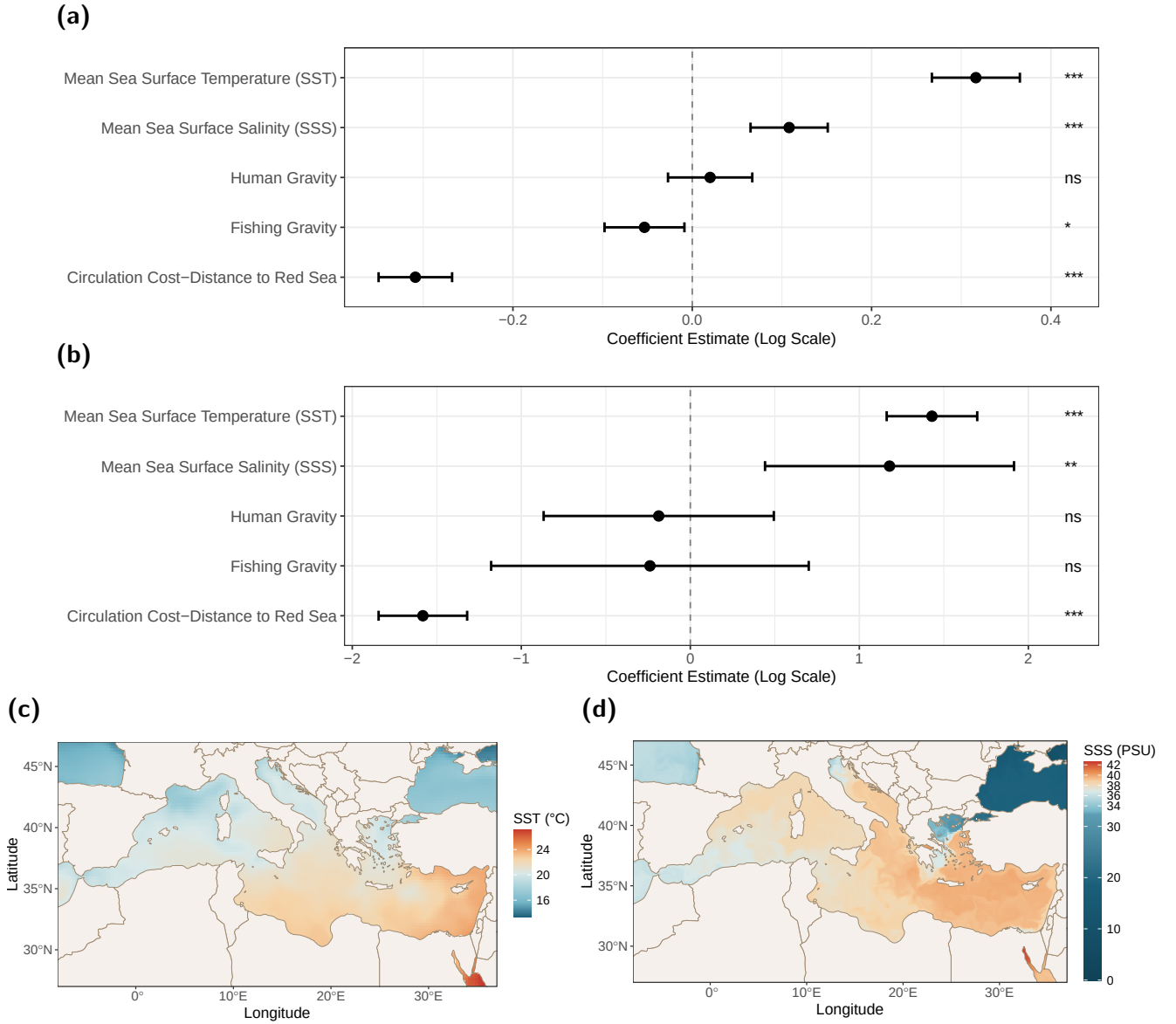
**Figure 1.** (a) A map of the Mediterranean Sea showing the 25 sampling locations of environmental DNA (eDNA) depicted as black dots. Coastal bathymetry, up to 200 m, is indicated in a lighter blue. Major circulation and currents (*sensu* Millot and Taupier-Letage, 2005) are illustrated as arrows according to the legend, while connections to adjacent oceans, seas, and their straits are written in blue. Additionally, written in green are the ecoregions (*sensu* Spalding et al., 2007) where the Western Mediterranean consists of the Balearic Sea, Ligurian Sea, and Tyrrhenian Sea, the Central Mediterranean consists of the Adriatic Sea and Ionian Sea, and the Eastern Mediterranean consists of the Aegean Sea and Levantine Sea. (b) Non-native fish species richness at each marine protected area (MPA). Points are interpolated using inverse distance weighting (power = 2) onto a 0.1° resolution grid to visualise the regional gradient. (c) and (d) Richness and number of eDNA reads, respectively, of non-native fish species as scatterplots (for individual eDNA reads per species see Supporting Figures 1 to 5). Each with a fitted local regression (LOESS) curve according to the longitude of MPAs in relation to the origin of non-native species.



**Figure 2.** Comparison of collected environmental DNA (eDNA) samples with an independent database of non-native fish species occurrences (Azzurro et al., 2026). (a) Box plots for distances between samples and observations from the non-native database. (b), (c), and (d) Maps of the Mediterranean Sea for selected species (silhouettes: Boulanger, 2022), showing in black the collected samples and, within a 200 km radius, in colour, the observations from the non-native database (see Supporting Figures 6 to 10 for the maps of all species).



**Figure 3.** Principal coordinates analysis (PCoA) using a Jaccard matrix for the beta diversity of fish species in each sampled marine protected area (MPA) and an indication of these points on a map of the Mediterranean Sea, with MPAs with no non-native fish species in grey and with ecoregions indicated (*sensu* Spalding et al., 2007). (a) and (b) Show global dissimilarity ( $\beta_{jac}$ ), comprising both turnover ( $\beta_{jtu}$ ) and nestedness ( $\beta_{jnes}$ ) components, whereas (c) and (d) show only the turnover component ( $\beta_{jtu}$ ).



**Figure 4.** Confidence plots, with a 95% confidence interval, for generalised linear mixed models (GLMMs) that test non-native fish species richness against environmental and human-pressure predictors, with (a) for the independent database of non-native fish species occurrences (Azzurro et al., 2026) and (b) for collected environmental DNA (eDNA) samples in the Mediterranean Sea. Non-significance levels are indicated as follows: \*\*\* =  $p < 0.001$ , \*\* =  $p < 0.01$ , \* =  $p < 0.05$ , and ns = non-significant. The GLMM uses a Poisson distribution, with geographic groupings for (a) coordinate groups and for (b) marine protected areas (MPAs) as a random effect, as well as the predictor variables as fixed effects. Data for variables, sea surface temperature (SST) and sea surface salinity (SSS), were extracted using the NOAA OI SST V2 High Resolution Dataset and E.U. Copernicus Marine Service Information ("Global Ocean Physics Analysis and Forecast", 2026; Huang et al., 2021). Furthermore, anthropogenic variables were used, namely fishing information (Coll et al., 2010) and human gravity, which was calculated following Caldwell et al., 2024; Cinner et al., 2018. Additionally, sea currents, also extracted using E.U. Copernicus Marine Service Information ("Global Total (COPERNICUS-GLOBCURRENT), Ekman and Geostrophic currents at the Surface and 15m", 2024), were used as a circulation variable. (c) Average Sea Surface Temperatures (SST) in the Mediterranean Sea using NOAA OI SST V2 High Resolution Dataset (Huang et al., 2021). (d) Average Sea Surface Salinity (SSS) in the Mediterranean Sea, given in Practical Salinity Units (PSU), using E.U. Copernicus Marine Service Information ("Global Ocean Physics Analysis and Forecast", 2026).